ERROR ESTIMATES OF THE CRANK-NICOLSON SCHEME FOR SOLVING BACKWARD STOCHASTIC DIFFERENTIAL EQUATIONS

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Abstract. In this paper, we study error estimates of a special case of the θ-scheme, the Crank-Nicolson (C-N) scheme, for solving the backward stochastic differential equation $-dy_t = f(t, y_t, z_t)dt - z_t dW_t$. We rigorously prove that under certain reasonable regularity conditions on the generator function f and the terminal condition y_T, the C-N scheme is of second-order accuracy for solving both y_t and z_t when errors are measured in the L^p $(p \geq 1)$ norm.

Key words. Backward stochastic differential equations, Crank-Nicolson scheme, θ-scheme, error estimate

AMS subject classifications. 60H35, 60H10, 65C30

1. Introduction.

Let $(\Omega, \mathcal{F}, P)$ be a probability space, $T > 0$ a finite time, and $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ a filtration satisfying usual conditions. Let $(\Omega, \mathcal{F}, \mathbb{F}, P)$ be a complete, filtered probability space on which a standard d-dimensional Brownian motion W_t is defined and $\mathcal{F}_0$ contains all the P-null sets of $\mathcal{F}$. Let $L^2 = L^2_{\mathcal{F}}(0, T)$ be the set of all $\mathcal{F}_t$-adapted and mean-square-integrable vector/matrix processes. We consider the backward stochastic differential equation (BSDE)

$$-dy_t = f(t, y_t, z_t)dt - z_t dW_t, \quad \forall\, t \in [0, T), \tag{1.1}$$

with the terminal condition

$$y_T = \xi,$$

where the generator $f(t, y_t, z_t) : \Omega \times [0, T] \times R^m \times R^{m \times d} \to R^m$ is $\mathcal{F}_t$-adapted for each (y_t, z_t), and the terminal variable $\xi \in L^2$ is $\mathcal{F}_T$-measurable [17]. Rewriting the BSDE (1.1) in the integral form gives us

$$y_t = \xi + \int_t^T f(s, y_s, z_s)\, ds - \int_t^T z_s\, dW_s, \quad \forall\, t \in [0, T]. \tag{1.2}$$

We note that the second integral term on the right-hand side of (1.2) is an Itô-type integral. A process (y_t, z_t): $[0, T] \times \Omega \to R^m \times R^{m \times d}$ is called a L^2-solution of the BSDE (1.2) if, in the probability space $(\Omega, \mathcal{F}, P)$, it is $\{\mathcal{F}_t\}$-adapted, square integrable, and satisfies the integral equation (1.2) [17]. In this paper, we assume that the BSDE (1.2) has a unique solution.

In 1990 Pardoux and Peng first proved in [17] existence and uniqueness of the solution of general nonlinear BSDEs (i.e, f is nonlinear), and later in [18], obtained some relations between BSDEs and stochastic partial differential equations (SPDEs). Since then, the theory of BSDEs has been extensively studied by many researchers and BSDEs have found applications in many fields, such as finance, risk measure,

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stochastic control, and etc. Peng also obtained the relation between BSDEs and parabolic PDEs in [20], and then the generalized stochastic maximum principle and the dynamic programming principle for stochastic control problems based on BSDEs in [19, 22]. The nonlinear $g-$expectation via a particular nonlinear BSDE was introduced in [21], and in [7] it was found that a dynamic coherent risk measure can be represented by a properly defined $g-$expectation. Thus, it is very important and useful to study solutions of BSDEs.

It is often very difficult to obtain analytic solutions of BSDEs, thus computing their numerical approximate solutions becomes highly desired. Based on the relation between the BSDE and the corresponding parabolic PDE, some numerical algorithms were proposed to solve BSDEs [3, 12, 13, 14, 15, 16, 20, 25], and furthermore, a four-step algorithm was proposed in [11] to solve a class of furtherly coupled equations called forward-backward stochastic differential equations (FBSDEs). In [26], a family of θ-schemes were proposed for solving general BSDEs. In particular, a special case of the θ-scheme – the Crank-Nicolson (C-N) scheme was *numerically* demonstrated to be second-order accurate. This accuracy result was theoretically proven in [23, 27] only for a family of very simplified BSDEs where the generator f is independent of z_t in (1.2), and a proof for BSDEs with general generators remains open till now. Some multi-step schemes were recently developed in [28] based on the Lagrange interpolation and the Gauss-Hermite quadratures. Accuracies of these multi-step schemes were numerically shown to be of high-order(1.2) for solving BSDEs, but again the result is only theoretically confirmed for BSDEs with a generator independent of z_t. There are also some other numerical methods for solving BSDEs (or FBSDEs), which were proposed based on directly discretizing BSDEs or FBSDEs, see [1, 2, 4, 5, 8, 9, 22, 24, 25] and references cited therein.

The aim of this paper is to study error estimates of the Crank-Nicolson scheme for solving BSDEs with a general generator f(1.2). We note the error estimate techniques used in this paper are quite different from standard approaches in the literature such as [1, 2, 4, 5, 8, 9, 22, 24, 25]. By combination of stochastic and deterministic analysis techniques and cancelations of truncation errors between successive time steps, we successfully obtain some optimal error estimates of the Crank-Nicolson scheme.

Some useful notations are first presented in the following:

- $\|X\|_{L^p}$ $(p \geq 1)$: the L^p-norm for $X \in L^p$ defined by $\mathbb{E}[|X|^p]^{\frac{1}{p}}$.
- $C_b^{l,k,k}$: the set of continuously differentiable functions $\phi : [0,T]\times R^d \times R^{m\times d} \to R$ with uniformly bounded partial derivatives $\partial_t^{l_1}\phi$ and $\partial_y^{k_1}\partial_z^{k_2}\phi$ for $\frac{1}{2} \leq l_1 \leq l$ and $1 \leq k_1 + k_2 \leq k$. $C_b^{l,k}$ is defined similarly.
- C_b^k: the set of functions $\psi : x \in R^d \to R$ with uniformly bounded partial derivatives $\partial_x^{k_1}\psi$ for $1 \leq k_1 \leq k$.
- $\mathcal{F}_s^{t,x}(t \leq s \leq T)$: the σ-field generated by the Brownian motion $\{x + W_r - W_t, t \leq r \leq s\}$ starting from the time-space point (t,x). Let $\mathcal{F}^{t,x} = \mathcal{F}_T^{t,x}$.
- $\mathbb{E}[X]$: the mathematical expectation of the random variable X.
- $\mathbb{E}_s^{t,x}[X]$: the conditional mathematical expectation of the random variable X under the σ-field $\mathcal{F}_s^{t,x}(t \leq s \leq T)$, that is $\mathbb{E}_s^{t,x}[X] = \mathbb{E}[X|\mathcal{F}_s^{t,x}]$. Let $\mathbb{E}_t^x[X] = \mathbb{E}[X|\mathcal{F}_t^{t,x}]$.
- $\partial_x\psi$: the matrix valued function $\partial_x\psi = (\partial_{x^j}\psi^i)_{m\times d}\,(1 \leq i \leq m, 1 \leq j \leq d)$ for vector function $\psi = (\psi^1, \ldots, \psi^m)^\top$.
- C: a generic positive constant, and may be different from line to line.

The rest of the paper is organized as follows. In Section 2, we briefly review the θ-scheme proposed in [26] for solving the BSDE (1.2) and one of its special form – the

Crank-Nicolson scheme. Then we rigorously derive error estimates of the C-N scheme in Section 3. Under certain reasonable regularity conditions on the generator f and the terminal condition y_T, we prove that the C-N scheme is second-order accurate when errors are measured in the L^p-norm ($p \geq 1$). Some concluding remarks are finally given in Section 4.

2. Review of the θ-Scheme and the Crank-Nicolson Scheme. In this section, we give a brief review of the θ-scheme proposed in [26]. For the time interval $[0,T]$, let us introduce the following partition

$$0 = t_0 < \cdots < t_N = T$$

with $\Delta t_n = t_{n+1} - t_n$, $n = 0, 1, \ldots, N-1$. Denote by (t_n, x^n) the time-space points.

2.1. Reference equations and the θ-Scheme. Let (y_t, z_t) be the solution of the BSDE (1.2). Then, for $0 \leq n \leq N-1$, it is easy to obtain

$$y_{t_n} = y_{t_{n+1}} + \int_{t_n}^{t_{n+1}} f(s, y_s, z_s)\, ds - \int_{t_n}^{t_{n+1}} z_s\, dW_s. \tag{2.1}$$

Taking the conditional mathematical expectation $\mathbb{E}_{t_n}^{x^n}[\cdot]$ on both sides of (2.1), we get

$$y_{t_n} = \mathbb{E}_{t_n}^{x^n}[y_{t_{n+1}}] + \int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^{x^n}[f_s]\, ds. \tag{2.2}$$

The integrand $\mathbb{E}_{t_n}^{x^n}[f_s]$ on the right-hand side of (2.2) is a deterministic smooth function of time s. We can use the following rule to approximate the integral in (2.2):

$$\int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^{x^n}[f_s]\, ds = (1-\theta_1)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f_{t_{n+1}}] + \theta_1 \Delta t_n f_{t_n} + R_y^n, \tag{2.3}$$

where the parameter $\theta_1 \in [0,1]$ and

$$R_y^n = \int_{t_n}^{t_{n+1}} \{\mathbb{E}_{t_n}^{x^n}[f_s] - (1-\theta_1)\mathbb{E}_{t_n}^{x^n}[f_{t_{n+1}}] - \theta_1 f_{t_n}\}\, ds.$$

Inserting (2.3) into (2.2), we get the first reference equation

$$y_{t_n} = \mathbb{E}_{t_n}^{x^n}[y_{t_{n+1}}] + \theta_1 \Delta t_n f_{t_n} + (1-\theta_1)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f_{t_{n+1}}] + R_y^n. \tag{2.4}$$

Let $\Delta_{t_n} W_s = W_s - W_{t_n}$ for $t_n \leq s \leq t_{n+1}$. Then $\Delta_{t_n} W_s$ is a standard Brownian motion with mean zero and variance $\sqrt{s - t_n}$. Multiplying (2.1) by $\Delta_{t_n} W_{t_{n+1}}^\top$, and then taking the conditional mathematical expectation $\mathbb{E}_{t_n}^{x^n}[\cdot]$ on both sides of the derived equation, we obtain by the Itô isometry formula

$$-\mathbb{E}_{t_n}^{x^n}[y_{t_{n+1}} \Delta_{t_n} W_{t_{n+1}}^\top] = \int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^{x^n}[f_s \Delta_{t_n} W_s^\top]\, ds - \int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^{x^n}[z_s]\, ds. \tag{2.5}$$

Following similar derivation of the equation (2.4) and using the fact $\Delta_{t_n} W_{t_n} = 0$, we obtain the second reference equation as

$$\begin{aligned} -\mathbb{E}_{t_n}^{x^n}[y_{t_{n+1}} \Delta_{t_n} W_{t_{n+1}}^\top] = {} & (1-\theta_2)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f_{t_{n+1}} \Delta_{t_n} W_{t_{n+1}}^\top] \\ & - \{(1-\theta_3)\Delta t_n \mathbb{E}_{t_n}^{x^n}[z_{t_{n+1}}] + \theta_3 \Delta t_n z_{t_n}\} + R_z^n, \end{aligned} \tag{2.6}$$

where

$$R_z^n = \int_{t_n}^{t_{n+1}} \mathbb{E}_{t_n}^{x^n}[f_s \Delta_{t_n} W_s^\top] ds - (1-\theta_2)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f_{t_{n+1}} \Delta_{t_n} W_{t_{n+1}}^\top]$$
$$- \int_{t_n}^{t_{n+1}} \{\mathbb{E}_{t_n}^{x^n}[z_s] ds - (1-\theta_3)\mathbb{E}_{t_n}^{x^n}[z_{t_{n+1}}] - \theta_3 z_{t_n}\} \, ds.$$

Define $\Delta W_{n+1} = W_{t_{n+1}} - W_{t_n}$ for $n = N-1, \ldots, 1, 0$. Clearly $\Delta W_{n+1} = \Delta_{t_n} W_{t_{n+1}}$. Based on the two reference equations (2.4) and (2.6), the following so-called θ-scheme was proposed for solving BSDEs in [26]: for $n = N-1, N-2, \ldots, 0$,

$$\begin{aligned} y^n &= \mathbb{E}_{t_n}^{x^n}[y^{n+1}] + \theta_1 \Delta t_n f(t_n, y^n, z^n) \\ &\quad + (1-\theta_1)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f(t_{n+1}, y^{n+1}, z^{n+1})], \qquad (2.7) \\ -\mathbb{E}_{t_n}^{x^n}[y^{n+1}\Delta W_{n+1}^\top] &= (1-\theta_2)\Delta t_n \mathbb{E}_{t_n}^{x^n}[f(t_{n+1}, y^{n+1}, z^{n+1})\Delta W_{n+1}^\top] \\ &\quad - \{(1-\theta_3)\Delta t_n \mathbb{E}_{t_n}^{x^n}[z^{n+1}] + \theta_3 \Delta t_n z^n\}, \qquad (2.8) \end{aligned}$$

where (y^n, z^n) is an approximation of (y_t, z_t) at time t_n. Note that (2.7) and (2.8) give us a time semi-discretization scheme. For further fully discretization in the space and numerical calculation of $\mathbb{E}_{t_n}^{x^n}[\cdot]$, please see [26, 28] for details.

2.2. The Crank-Nicolson scheme. Notice that at time $t_N = T$ we only have the information $y_{t_N} = \xi$ but z_{t_N} is out of our knowledge. Thus in order to proceed the θ-scheme (2.7) and (2.8), we need an initialization process by choosing $\theta_1 = \theta_2 = \theta_3 = 1$ in (2.7) and (2.8) at $n = N-1$ such that we can solve y^{N-1} and z^{N-1} from $y^N = y_{t_N}$ by

$$y^{N-1} = \mathbb{E}_{t_{N-1}}^{x^{N-1}}[y^N] + \Delta t_{N-1} f(t_{N-1}, y^{N-1}, z^{N-1}), \qquad (2.9)$$
$$\Delta t_{N-1} z^{N-1} = \mathbb{E}_{t_{N-1}}^{x^{N-1}}[y^N \Delta W_N^\top]. \qquad (2.10)$$

In the following steps, by setting $\theta_1 = \theta_2 = \theta_3 = \frac{1}{2}$, we obtain a special case of the θ-scheme – the Crank-Nicolson (C-N) scheme for solving general BSDEs:

Given y^N which is an approximation to y_{t_N}.

(i) Solve (y^{N-1}, z^{N-1}) according to (2.9) and (2.10);

(ii) For $n = N-2, N-3, \ldots, 0$, solve (y^n, z^n) by

$$\begin{aligned} y^n &= \mathbb{E}_{t_n}^{x^n}[y^{n+1}] + \frac{1}{2}\Delta t_n f(t_n, y^n, z^n) \\ &\quad + \frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[f(t_{n+1}, y^{n+1}, z^{n+1})], \qquad (2.11) \\ \frac{1}{2}\Delta t_n z^n &= \mathbb{E}_{t_n}^{x^n}[y^{n+1}\Delta W_{n+1}^\top] - \frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[z^{n+1}] \\ &\quad + \frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[f(t_{n+1}, y^{n+1}, z^{n+1})\Delta W_{n+1}^\top]. \qquad (2.12) \end{aligned}$$

REMARK 1. *We note that the C-N scheme defined by (2.11) and (2.12) is explicit in solving z^n. Then the Lipchitz condition of f leads to that the C-N scheme has a unique solution (y^n, z^n) for small time partition step Δt.*

Note that the truncation errors of (2.11) and (2.12) in the step (ii) are one-order smaller (in time step size) than that of (2.9) and (2.10) in the initialization step (i).

Thus in order to obtain optimal error estimates of the C-N scheme, we also need the following assumption.

ASSUMPTION 1. *$\Delta t_n = \Delta t$ for $0 \le n \le N-2$ and $\Delta t_{N-1} = (\Delta t)^2$ with the constant $\Delta t > 0$.*

In the rest of the paper, we will always assume Assumption 1 holds.

REMARK 2. *Various experiments presented in [26, 27, 28] showed that the above C-N scheme is numerically second-order accurate. The authors in [23, 27] confined the discussions to a family of simplified BSDEs*

$$y_t = \varphi(W_T) + \int_t^T f(s, y_s)\, ds - \int_t^T z_s\, dW_s, \quad t \in [0, T], \tag{2.13}$$

(i.e, the generator f is independent of z_t), and obtained some theoretical results on convergence of the C-N scheme. In particular, it was rigorously proven based on a variation method that the C-N scheme is second-order accurate for this simplified case under certain regularity conditions. However, the analysis techniques used there can not be applied to BSDEs with a general generator $f(t, y_t, z_t)$.

3. Error estimates of the Crank-Nicolson Scheme. Suppose that the terminal condition is given by $y_T = \varphi(x_T)$, where x_t is a diffusion process with drift coefficient $b(t, x_t)$ and volatility coefficient $\sigma(t, x_t)$. Under some regularity conditions on f and φ, it was shown in [11, 12, 20] that the solution (y_t, z_t) of (1.2) can be represented as

$$y_t = u(t, x_t), \qquad z_t = \nabla_x u(t, x_t)\sigma(t, x_t), \qquad \forall\, t \in [0, T), \tag{3.1}$$

where $u(t, x)$ is the unique classic solution of the following parabolic PDE

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sum_{i,j=1}^{d} (\sigma\sigma^*)_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^{d} b_i \frac{\partial u}{\partial x_i} + f(t, u, \nabla_x u\sigma) = 0, \tag{3.2}$$

with the terminal condition $u(T, x) = \varphi(x)$, and $\nabla_x u$ is the gradient of u with respect to the spacial variable x.

REMARK 3. *The smoothness of u clearly depends on b, σ, φ and f (see [6, 10]). For examples, if b, σ are uniformly Lipschitz continuous w.r.t. x_t, f are uniformly Lipschitz continuous w.r.t. y_t and z_t, b, σ, f are all Hölder continuous of parameter $\frac{1}{2}$ w.r.t. t, and $\varphi \in C_b^{2+\alpha}$ for some $\alpha \in (0, 1)$, then $u \in C_b^{1,2}$; if $b, \sigma \in C_b^{3,4}$, $f \in C_b^{3,4,4}$ and $\varphi \in C_b^{5+\alpha}$ for some $\alpha \in (0, 1)$, then $u \in C_b^{3,5}$; if $b, \sigma \in C_b^{3,6}$, $f \in C_b^{3,6,6}$ and $\varphi \in C_b^{7+\alpha}$ for some $\alpha \in (0, 1)$, then $u \in C_b^{3,7}$.*

Without loss of generality, in the remainder of the paper we only consider the case of one-dimensional BSDEs (i.e., $m = d = 1$) and the diffusion process x_t being a standard Brownian motion (i.e., $x_t = W_t$, and in this case $\sigma = 1$ and $b = 0$), but we note that error estimates obtained in the sequel obviously also hold for more general settings.

3.1. Some important lemmas. Define $\Delta_{t_i}^{x^i} W_s = x^i + W_s - W_{t_i}$ for $t_i \le s \le t_{i+1}$. Clearly $x^{n+1} = x^n + W_{t_{n+1}} - W_{t_n}$ for $n = 0, 1, \ldots, N-1$. First we introduce the following lemma.

LEMMA 3.1. *If $H \in C_b^{3,5}$, then when Δt is sufficiently small it holds that for $1 \leq i \leq N-2$,*

$$\left|\mathbb{E}_{t_{i-1}}^{x^{i-1}}\left[\Delta W_i \int_{t_i}^{t_{i+1}} \left\{H(t, \Delta_{t_i}^{x^i} W_t) - \frac{H(t_i, x^i) + H(t_{i+1}, \Delta_{t_i}^{x^i} W_{t_{i+1}})}{2}\right\} dt\right]\right| \leq C(\Delta t)^4, \tag{3.3}$$

where $C > 0$ is a generic constant depending only on the bounds of derivatives of the function H.

Proof. Since ΔW_i is $\mathcal{F}_{t_i}$-measurable, we have the identity

$$\begin{aligned}
&\mathbb{E}_{t_{i-1}}^{x^{i-1}}\Big[\Delta W_i \int_{t_i}^{t_{i+1}} \Big\{H(t, \Delta_{t_i}^{x^i} W_t) - \frac{H(t_i, x^i) + H(t_{i+1}, \Delta_{t_i}^{x^i} W_{t_{i+1}})}{2}\Big\} dt\Big] \\
&\quad = \mathbb{E}_{t_{i-1}}^{x^{i-1}}\Big[\Delta W_i \mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}} \Big\{H(t, \Delta_{t_i}^{x^i} W_t) - \frac{H(t_i, x^i) + H(t_{i+1}, \Delta_{t_i}^{x^i} W_{t_{i+1}})}{2}\Big\} dt\Big]\Big].
\end{aligned} \tag{3.4}$$

By the Itô formula, we have

$$\begin{aligned}
H(t, \Delta_{t_i}^{x^i} W_t) = H(t_i, x^i) &+ \int_{t_i}^{t} \Big(H_t(s, \Delta_{t_i}^{x^i} W_s) + \frac{1}{2} H_{xx}(s, \Delta_{t_i}^{x^i} W_s)\Big) ds \\
&+ \int_{t_i}^{t} H_x(s, \Delta_{t_i}^{x^i} W_s)\, dW_s.
\end{aligned} \tag{3.5}$$

Notice that

$$\begin{aligned}
&H_t(s, \Delta_{t_i}^{x^i} W_s) + \frac{1}{2} H_{xx}(s, \Delta_{t_i}^{x^i} W_s) \\
&\quad = H_t(t_i, x^i) + \frac{1}{2} H_{xx}(t_i, x^i) \\
&\qquad + \int_{t_i}^{s} \Big(H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau) + H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau) + \frac{1}{4} H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)\Big)\, d\tau \\
&\qquad + \int_{t_i}^{s} \Big(H_{tx}(\tau, \Delta_{t_i}^{x^i} W_\tau) + \frac{1}{2} H_{xxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)\Big)\, dW_\tau.
\end{aligned} \tag{3.6}$$

By (3.5) and (3.6) we easily get

$$\begin{aligned}
&\mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}} H(t, \Delta_{t_i}^{x^i} W_t)\, dt\Big] \\
&\qquad = H(t_i, x^i)\Delta t + \frac{1}{2} H_t(t_i, x^i)(\Delta t)^2 + \frac{1}{4} H_{xx}(t_i, x^i)(\Delta t)^2 \\
&\qquad\quad + \int_{t_i}^{t_{i+1}} \int_{t_i}^{t} \int_{t_i}^{s} \Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau) + H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \\
&\qquad\qquad + \frac{1}{4} \mathbb{E}_{t_i}^{x^i}[H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)]\Big)\, d\tau ds dt,
\end{aligned} \tag{3.7}$$

and

$$\begin{aligned}
&\mathbb{E}_{t_i}^{x^i}[\int_{t_i}^{t_{i+1}} H(t_{i+1}, \Delta_{t_i}^{x^i} W_{t_{i+1}})\, dt] \\
&\qquad = H(t_i, x^i)\Delta t + H_t(t_i, x^i)(\Delta t)^2 + \frac{1}{2} H_{xx}(t_i, x^i)(\Delta t)^2 \\
&\qquad\quad + \int_{t_i}^{t_{i+1}} \int_{t_i}^{t_{i+1}} \int_{t_i}^{s} \Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \\
&\qquad\qquad + \mathbb{E}_{t_i}^{x^i}[H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau) + \frac{1}{4} H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \Big)\, d\tau ds dt.
\end{aligned} \tag{3.8}$$

Then by (3.7) and (3.8) we deduce

$$\begin{aligned}
&\mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}} \Big(H(t, \Delta_{t_i}^{x^i} W_t) - \frac{H(t_i, x^i) + H(t_{i+1}, \Delta_{t_i}^{x^i} W_{t_{i+1}})}{2} \Big)\, dt \Big] \\
&\qquad = \int_{t_i}^{t_{i+1}} \int_{t_i}^{t} \int_{t_i}^{s} \Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \\
&\qquad\qquad + \mathbb{E}_{t_i}^{x^i}[H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau) + \frac{1}{4} H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \Big)\, d\tau ds dt \\
&\qquad\quad - \frac{1}{2} \int_{t_i}^{t_{i+1}} \int_{t_i}^{t_{i+1}} \int_{t_i}^{s} \Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \\
&\qquad\qquad + \mathbb{E}_{t_i}^{x^i}[H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau) + \frac{1}{4} H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)] \Big)\, d\tau ds dt.
\end{aligned} \tag{3.9}$$

According to the Taylor expansion it holds

$$\begin{aligned}
H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau) =& H_{tt}(\tau, x^i) + H_{ttx}(\tau, x^i + \alpha_1(W_\tau - W_{t_i}))(W_\tau - W_{t_i}) \\
=& H_{tt}(\tau, x^{i-1}) + H_{ttx}(\tau, x^{i-1} + \alpha_2 \Delta W_{t_i})\Delta W_{t_i} \\
& + H_{ttx}(\tau, x^i + \alpha_1(W_\tau - W_{t_i}))(W_\tau - W_{t_i}),
\end{aligned} \tag{3.10}$$

where α_1 and α_2 are some positive numbers in $[0, 1]$. Then from the fact

$$\mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{tt}(\tau, x^{i-1})] = H_{tt}(\tau, x^{i-1}) \mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i] = 0,$$

we can get

$$|\mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{tt}(\tau, \Delta_{t_i}^{x^i} W_\tau)]| \le C\Delta t. \tag{3.11}$$

By using similar analysis and the facts

$$\mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{txx}(\tau, x^{i-1})] = 0, \quad \mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{xxxx}(\tau, x^{i-1})] = 0,$$

we also can obtain the following estimates:

$$\begin{aligned}
&|\mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{txx}(\tau, \Delta_{t_i}^{x^i} W_\tau)]| \le C\Delta t, \\
&|\mathbb{E}_{t_{i-1}}^{x^{i-1}}[\Delta W_i H_{xxxx}(\tau, \Delta_{t_i}^{x^i} W_\tau)]| \le C\Delta t.
\end{aligned} \tag{3.12}$$

Combination of (3.4), (3.9), (3.11) and (3.12) gives us the inequality (3.3). The proof is completed. ∎

LEMMA 3.2. *If $H \in C_b^{3,6}$, then when Δt is sufficiently small it holds that for $0 \le i \le N-3$,*

$$\Big|\mathbb{E}_{t_i}^{x^i}\Big[\mathbb{E}_{t_{i+1}}^{x^{i+1}}\Big[\int_{t_{i+1}}^{t_{i+2}}\{H(t,\Delta_{t_{i+1}}^{x^{i+1}}W_t)-\frac{H(t_{i+1},x^{i+1})+H(t_{i+2},\Delta_{t_{i+1}}^{x^{i+1}}W_{t_{i+2}})}{2}\Big\}\,dt\Big]\Big]$$
$$-\mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}}\Big\{H(t,\Delta_{t_i}^{x^i}W_t)-\frac{H(t_i,x^i)+H(t_{i+1},\Delta_{t_i}^{x^i}W_{t_{i+1}})}{2}\Big\}\,dt\Big]\Big| \le C(\Delta t)^4, \tag{3.13}$$

where $C>0$ is a generic constant depending only on the bounds of derivatives of the function H.

Proof. Similar to (3.9), we can obtain the following two inequalities:

$$\begin{aligned}
&\mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}}\Big(H(t,\Delta_{t_i}^{x^i}W_t)-\frac{H(t_i,x^i)-H(t_{i+1},\Delta_{t_i}^{x^i}W_{t_{i+1}})}{2}\Big)\,dt\Big]\\
&\quad=\int_{t_i}^{t_{i+1}}\int_{t_i}^{t}\int_{t_i}^{s}\Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau,\Delta_{t_i}^{x^i}W_\tau)]\\
&\qquad\qquad+\mathbb{E}_{t_i}^{x^i}[H_{txx}(\tau,\Delta_{t_i}^{x^i}W_\tau)+\frac{1}{4}H_{xxxx}(\tau,\Delta_{t_i}^{x^i}W_\tau)]\Big)\,d\tau ds dt\\
&\qquad-\frac{1}{2}\int_{t_i}^{t_{i+1}}\int_{t_i}^{t_{i+1}}\int_{t_i}^{s}\Big(\mathbb{E}_{t_i}^{x^i}[H_{tt}(\tau,\Delta_{t_i}^{x^i}W_\tau)]\\
&\qquad\qquad+\mathbb{E}_{t_i}^{x^i}[H_{txx}(\tau,\Delta_{t_i}^{x^i}W_\tau)+\frac{1}{4}H_{xxxx}(\tau,\Delta_{t_i}^{x^i}W_\tau)]\Big)\,d\tau ds dt,
\end{aligned}$$

and

$$\begin{aligned}
&\mathbb{E}_{t_{i+1}}^{x^{i+1}}\Big[\int_{t_{i+1}}^{t_{i+2}}\Big(H(t,\Delta_{t_{i+1}}^{x^{i+1}}W_t)-\frac{H(t_{i+1},\Delta_{t_{i+1}}^{x^{i+1}}W_{t_{i+1}})+H(t_{i+2},\Delta_{t_{i+1}}^{x^{i+1}}W_{t_{i+2}})}{2}\Big)\,dt\Big]\\
&\quad=\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{t}\int_{t_{i+1}}^{s}\Big(\mathbb{E}_{t_{i+1}}^{x^{i+1}}[H_{tt}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)]\\
&\qquad\qquad+\mathbb{E}_{t_{i+1}}^{x^{i+1}}[H_{txx}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)+\frac{1}{4}H_{xxxx}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)]\Big)\,d\tau ds dt\\
&\qquad-\frac{1}{2}\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{s}\Big(\mathbb{E}_{t_{i+1}}^{x^{i+1}}[H_{tt}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)]\\
&\qquad\qquad+\mathbb{E}_{t_{i+1}}^{x^{i+1}}[H_{txx}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)+\frac{1}{4}H_{xxxx}(\tau,\Delta_{t_{i+1}}^{x^{i+1}}W_\tau)]\Big)\,d\tau ds dt.
\end{aligned}$$

Since $x^{i+1}=x^i+W_{t_{i+1}}-W_{t_i}$, we have

$$\Delta_{t_{i+1}}^{x^{i+1}}W_\tau = x^{i+1}+W_\tau-W_{t_{i+1}}=\Delta_{t_i}^{x^i}W_\tau.$$

Then for $t_i \le \tau \le t_{i+2}$, by the Taylor expansion we have

$$\begin{aligned}
&H_{tt}(\tau,\Delta_{t_i}^{x^i}W_\tau)\\
&\quad=H_{tt}(t_i,x^i)+H_{ttt}(t_i+\alpha_1(\tau-t_i),x^i+\alpha_1(W_\tau-W_{t_i}))(\tau-t_i)\\
&\qquad+H_{ttx}(t_i+\alpha_1(\tau-t_i),x^i)(W_\tau-W_{t_i})\\
&\qquad+H_{ttxx}(t_i+\alpha_1(\tau-t_i),x^i+\alpha_2(W_\tau-W_{t_i}))\alpha_1(W_\tau-W_{t_i})^2,
\end{aligned}\tag{3.14}$$

$$
\begin{aligned}
&H_{txx}(\tau,\Delta_{t_i}^{x^i}W_\tau)\\
&\quad= H_{txx}(t_i,x^i)+H_{ttxx}(t_i+\alpha_3(\tau-t_i),x^i+\alpha_3(W_\tau-W_{t_i}))(\tau-t_i)\\
&\qquad+H_{txxx}(t_i+\alpha_3(\tau-t_i),x^i)(W_\tau-W_{t_i})\\
&\qquad+H_{txxxx}(t_i+\alpha_3(\tau-t_i),x^i+\alpha_4(W_\tau-W_{t_i}))\alpha_3(W_\tau-W_{t_i})^2,
\end{aligned}
\tag{3.15}
$$

$$
\begin{aligned}
&H_{xxxx}(\tau,\Delta_{t_i}^{x^i}W_\tau)\\
&= H_{xxxx}(t_i,x^i)+H_{txxxx}(t_i+\alpha_5(\tau-t_i),x^i+\alpha_5(W_\tau-W_{t_i}))(\tau-t_i)\\
&\quad+H_{xxxxx}(t_i+\alpha_5(\tau-t_i),x^i)(W_\tau-W_{t_i})\\
&\quad+H_{xxxxxx}(t_i+\alpha_5(\tau-t_i),x^i+\alpha_6(W_\tau-W_{t_i}))\alpha_5(W_\tau-W_{t_i})^2,
\end{aligned}
\tag{3.16}
$$

where $\alpha_i(i=1,2,\ldots,6)$ are some positive numbers in $[0,1]$. Notice that $H_{ttx}(t_i+\alpha_1(\tau-t_i),x^i)$, $H_{txxx}(t_i+\alpha_3(\tau-t_i),x^i)$, $H_{xxxxx}(t_i+\alpha_5(\tau-t_i),x^i)$ are all $\mathcal{F}_{t_i}$-measurable, thus we have

$$
\begin{aligned}
&\mathbb{E}_{t_i}^{x^i}[H_{ttx}(t_i+\alpha_1(\tau-t_i),x^i)(W_\tau-W_{t_i})]=0,\\
&\mathbb{E}_{t_i}^{x^i}[H_{txxx}(t_i+\alpha_3(\tau-t_i),x^i)(W_\tau-W_{t_i})]=0,\\
&\mathbb{E}_{t_i}^{x^i}[H_{xxxxx}(t_i+\alpha_5(\tau-t_i),x^i)(W_\tau-W_{t_i})]=0.
\end{aligned}
$$

Now under the conditions of the lemma and by the equations (3.14), (3.15) and (3.16), we deduce

$$
\begin{aligned}
&\Big|\mathbb{E}_{t_i}^{x^i}\Big[\mathbb{E}_{t_{i+1}}^{x^{i+1}}\Big[\int_{t_{i+1}}^{t_{i+2}}\Big\{H(t,\Delta_{t_i}^{x^i}W_t)-\frac{H(t_{i+1},\Delta_{t_i}^{x^i}W_{t_{i+1}})+H(t_{i+2},\Delta_{t_i}^{x^i}W_{t_{i+2}})}{2}\Big\}\,dt\Big]\Big]\\
&\qquad-\mathbb{E}_{t_i}^{x^i}\Big[\int_{t_i}^{t_{i+1}}\Big\{H(t,\Delta_{t_i}^{x^i}W_t)-\frac{H(t_i,x^i)+H(t_{i+1},\Delta_{t_i}^{x^i}W_{t_{i+1}})}{2}\Big\}\,dt\Big]\Big|\\
&\quad\le\Big|\int_{t_i}^{t_{i+1}}\int_{t_i}^{t}\int_{t_i}^{s}\Big(H_{tt}(t_i,x^i)+H_{txx}(t_i,x^i)+\frac14 H_{xxxx}(t_i,x^i)\Big)\,d\tau ds dt\\
&\qquad-\frac12\int_{t_i}^{t_{i+1}}\int_{t_i}^{t_{i+1}}\int_{t_i}^{s}\Big(H_{tt}(t_i,x^i)+H_{txx}(t_i,x^i)+\frac14 H_{xxxx}(t_i,x^i)\Big)\,d\tau ds dt\\
&\qquad-\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{t}\int_{t_{i+1}}^{s}\Big(H_{tt}(t_i,x^i)+H_{txx}(t_i,x^i)+\frac14 H_{xxxx}(t_i,x^i)\Big)\,d\tau ds dt\\
&\qquad+\frac12\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{t_{i+2}}\int_{t_{i+1}}^{s}\Big(H_{tt}(t_i,x^i)+H_{txx}(t_i,x^i)+\frac14 H_{xxxx}(t_i,x^i)\Big)\,d\tau ds dt\Big|\\
&\qquad+C(\Delta t)^4\\
&\quad=\Big(\frac16-\frac14-\frac16+\frac14\Big)(\Delta t)^3\Big|H_{tt}(t_i,x^i)+H_{txx}(t_i,x^i)+\frac14 H_{xxxx}(t_i,x^i)\Big|+C(\Delta t)^4\\
&\quad\le C(\Delta t)^4,
\end{aligned}
$$

where C is a constant which depends only on bounds of the derivatives of H. □

Upper bounds of $|R_y^n|$ and $|R_z^n|$ defined in (2.4) and (2.6) are presented in the following.

LEMMA 3.3. *Let (y_t, z_t) is the solution of (1.2), and let R_y^n and R_z^n be the truncation errors defined in (2.4) and (2.6) for the θ-scheme. Suppose Δt is sufficiently small.*

1. *If the terminal function $\varphi \in C_b^{3+\alpha}$ for some $\alpha \in (0,1)$ and the generator function $f \in C_b^{1,2,2}$, then it holds that for $0 \leq n < N-1$*
$$|R_y^n| \leq C(\Delta t_n)^2, \quad |R_z^n| \leq C(\Delta t_n)^2. \tag{3.17}$$
2. *In particular, for the Crank-Nicolson scheme ($\theta_i = 1/2,\ i = 1,2,3$), if $\varphi \in C_b^{5+\alpha}$ for some $\alpha \in (0,1)$ and $f \in C_b^{2,4,4}$, then it holds that for $0 \leq n < N-1$*
$$|R_y^n| \leq C(\Delta t_n)^3, \quad |R_z^n| \leq C(\Delta t_n)^3. \tag{3.18}$$

Note here $C > 0$ is a positive constant depending only on T and the bounds of the derivatives of the functions φ and f.

Under the conditions of f and φ in Lemma 3.3, using the Taylor expansion and the properties of the Brownian motion W_t, it is easy to prove Lemma 3.3, see [23, 27] for details.

LEMMA 3.4. *Let (y_t, z_t) be the solution of (1.2), and R_y^i and R_z^i be the truncation errors defined in (2.4) and (2.6) for the C-N scheme ($\theta_i = 1/2,\ i = 1,2,3$), i.e.,*
$$R_y^i = \int_{t_i}^{t_{i+1}} \{\mathbb{E}_{t_i}^{x^i}[f(s, y_s, z_s)] - \frac{1}{2}\mathbb{E}_{t_i}^{x^i}[f(t_{i+1}, y_{t_{i+1}}, z_{t_{i+1}})] - \frac{1}{2} f(t_i, y_{t_i}, z_{t_i})\}\, ds, \tag{3.19}$$
$$\begin{aligned} R_z^i = &\int_{t_i}^{t_{i+1}} \mathbb{E}_{t_i}^{x^i}[f(s, y_s, z_s)\Delta_{t_i} W_s^\top] ds - \frac{1}{2}\Delta t_i \mathbb{E}_{t_i}^{x^i}[f(t_{i+1}, y_{t_{i+1}}, z_{t_{i+1}})\Delta_{t_i} W_{t_{i+1}}^\top] \\ &- \int_{t_i}^{t_{i+1}} \{\mathbb{E}_{t_i}^{x^i}[z_s] - \frac{1}{2}\mathbb{E}_{t_i}^{x^i}[z_{t_{i+1}}] - \frac{1}{2} z_{t_i}\}\, ds, \end{aligned} \tag{3.20}$$
for $0 \leq i \leq N-1$. If $\varphi \in C_b^{6+\alpha}$ for some $\alpha \in (0,1)$ and $f \in C_b^{3,5,5}$, then when Δt is sufficiently small, we have
$$|\mathbb{E}_{t_i}^{x^i}[R_y^{i+1}\Delta W_{i+1}]| \leq C(\Delta t)^4 \tag{3.21}$$
for $0 \leq i \leq N-3$. And if $\varphi \in C_b^{7+\alpha}$ for some $\alpha \in (0,1)$ and $f \in C_b^{3,6,6}$, then when Δt is sufficiently small it holds that for $0 \leq i \leq N-3$,
$$|R_z^i - \mathbb{E}_{t_i}^{x^i}[R_z^{i+1}]| \leq C(\Delta t)^4. \tag{3.22}$$
Here C is a positive constant depending only on T and the bounds of derivatives of functions φ and f .

Proof. Under the conditions of the lemma, the solution (y_t, z_t) of the BSDE (1.2) can be represented as
$$y_t = u(t, W_t), \quad z_t = \nabla_x u(t, W_t), \qquad \forall\, t \in [0, T), \tag{3.23}$$
where $u(t,x)$ satisfies the parabolic PDE (3.1). Let $H(t, W_t) = f(t, y_t, z_t)$, then $H(t, W_t) = f(t, u(t, W_t), \nabla_x u(t, W_t))$ according to (3.1). By the theory of partial differential equations [6], it is easy to check that the functions H satisfy the conditions in Lemma 3.1, thus we can easily get the estimates (3.21) using Lemma 3.1.

Similarly, by letting
$$\begin{aligned} H(t, W_t) &= f(t, y_t, z_t)\Delta_{t_i} W_t - z_t \\ &= f(t, u(t, W_t), \nabla_x u(t, W_t))\Delta_{t_i} W_t - \nabla_x u(t, W_t) \end{aligned}$$
and using Lemma 3.2, we can get (3.22). The proof is completed. □

3.2. Error estimates. Let (y_t, z_t) be the solution of the BSDE (1.2) with the terminal condition $y_T = \varphi(W_T)$, and (y^n, z^n) be its approximate solution produced by using the Crank-Nicolson scheme. For $0 \le n \le N$, let

$$e_y^n = y_{t_n} - y^n, \quad e_z^n = z_{t_n} - z^n, \quad e_f^n = f(t_n, y_{t_n}, z_{t_n}) - f(t_n, y^n, z^n).$$

By the equations (2.4), (2.6), (2.9) and (2.10), we get that for $n = N-1$,

$$e_y^{N-1} = \mathbb{E}_{t_{N-1}}^{x^{N-1}}[e_y^N] + \Delta t_{N-1} e_f^{N-1} + R_y^{N-1}, \tag{3.24}$$

and

$$\Delta t_{N-1} e_z^{N-1} = \mathbb{E}_{t_{N-1}}^{x^{N-1}}[e_y^N \Delta W_N] + R_z^{N-1}. \tag{3.25}$$

Then using (2.4), (2.6), (2.11) and (2.12), we have that for $0 \le n \le N-2$,

$$e_y^n = \mathbb{E}_{t_n}^{x^n}[e_y^{n+1}] + \frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[e_f^{n+1}] + \frac{1}{2}\Delta t_n e_f^n + R_y^n, \tag{3.26}$$

and

$$\begin{aligned} \frac{1}{2}\Delta t_n e_z^n = & -\frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[e_z^{n+1}] + \mathbb{E}_{t_n}^{x^n}[e_y^{n+1}\Delta W_{n+1}] \\ & + \frac{1}{2}\Delta t_n \mathbb{E}_{t_n}^{x^n}[e_f^{n+1}\Delta W_{n+1}] + R_z^n. \end{aligned} \tag{3.27}$$

We also note that $\Delta t_{N-1} = (\Delta t)^2$ and $\Delta t_n = \Delta t$ for $0 \le n \le N-2$ according to Assumption 1.

LEMMA 3.5. *Suppose that $f \in C_b^{3,4,4}$, $\varphi \in C_b^{5+\alpha}$ for some $\alpha \in (0,1)$, and the initial error satisfies*

$$\mathbb{E}_{t_{N-1}}^{x^{N-1}}[|y_{t_N} - y^N|^2] \le C(\Delta t_{N-1})^3 \tag{3.28}$$

for some constant $C > 0$. Then when Δt is sufficiently small, it holds that for $N-3 \le n \le N-1$,

$$|e_y^n| \le C(\Delta t)^3, \quad |e_z^n| \le C(\Delta t)^2, \tag{3.29}$$

where $C > 0$ is a generic constant depending only on T and the bounds of derivatives of φ and f.

Proof. According to Lemma 3.3, we know that

$$|R_y^{N-1}| \le C(\Delta t_{N-1})^2, \qquad |R_z^{N-1}| \le C(\Delta t_{N-1})^2. \tag{3.30}$$

Let L be the Lipschitz constant of f with respect to y and z. Recall that $\Delta t_{N-1} = (\Delta t)^2$. By the inequality (3.25), (3.28) and (3.30) and Hölder's inequality, we deduce

$$\begin{aligned} |e_z^{N-1}| &\le \frac{1}{\Delta t_{N-1}}\left(|\mathbb{E}_{t_{N-1}}^{x^{N-1}}[e_y^N \Delta W_N]| + |R_z^{N-1}|\right) \\ &\le \frac{1}{\Delta t_{N-1}} \mathbb{E}_{t_{N-1}}^{x^{N-1}}[|e_y^N|^2]^{\frac{1}{2}} \mathbb{E}_{t_{N-1}}^{x^{N-1}}[|\Delta W_N|^2]^{\frac{1}{2}} + \frac{|R_z^{N-1}|}{\Delta t_{N-1}} \\ &\le \frac{1}{\Delta t_{N-1}} C^{1/2}(\Delta t_{N-1})^{3/2}(\Delta t_{N-1})^{1/2} + C\Delta t_{N-1} \\ &\le C\Delta t_{N-1} = C(\Delta t)^2, \end{aligned} \tag{3.31}$$

and

$$
\begin{aligned}
|e_y^{N-1}| &\leq \mathbb{E}_{t_{N-1}}^{x^{N-1}}[|e_y^N|] + L\Delta t_{N-1}(|e_y^{N-1}| + |e_z^{N-1}|) + |R_y^{N-1}| \\
&\leq \mathbb{E}_{t_{N-1}}^{x^{N-1}}[|e_y^N|^2]^{\frac{1}{2}} + L\Delta t_{N-1}|e_y^{N-1}| + LC(\Delta t_{N-1})^2 + C(\Delta t_{N-1})^2 \\
&\leq C^{1/2}(\Delta t_{N-1})^{3/2} + L\Delta t_{N-1}|e_y^{N-1}| + (L+1)C(\Delta t_{N-1})^2 \\
&\leq L\Delta t_{N-1}|e_y^{N-1}| + C(\Delta t_{N-1})^{3/2}.
\end{aligned}
$$

Then we get

$$
|e_y^{N-1}| \leq \frac{C(\Delta t_{N-1})^{3/2}}{1 - L\Delta t_{N-1}} \leq C(\Delta t_{N-1})^{\frac{3}{2}} = C(\Delta t)^3. \tag{3.32}
$$

Recall that $\Delta t_n = \Delta t$ for $0 \leq n \leq N-2$. By Lemma 3.3, we have that for $0 \leq n \leq N-2$,

$$
|R_y^n| \leq C(\Delta t)^3, \qquad |R_z^n| \leq C(\Delta t)^3. \tag{3.33}
$$

Now we estimate e_y^n and e_z^n for $n = N-2, N-3$. From the equation (3.27), the inequalities (3.31), (3.32) and (3.33), we deduce by the Hölder inequality that

$$
\begin{aligned}
|e_z^{N-2}| &\leq \mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_z^{N-1}|] + \frac{2}{\Delta t}\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_y^{N-1}|^2]^{\frac{1}{2}}\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|\Delta W_{N-1}|^2]^{\frac{1}{2}} \\
&\qquad + \mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_f^{N-1}|^2]^{\frac{1}{2}}\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|\Delta W_{N-1}|^2]^{\frac{1}{2}} + \frac{2}{\Delta t}|R_z^{N-2}| \\
&\leq C(\Delta t)^2 + 2C(\Delta t)^{5/2} \\
&\qquad + 2L\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_y^{N-1}|^2 + |e_z^{N-1}|^2]^{\frac{1}{2}}\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|\Delta W_{N-1}|^2]^{\frac{1}{2}} + 2C(\Delta t)^2 \\
&\leq C(\Delta t)^2 + 2LC(\Delta t)^{5/2} \leq C(\Delta t)^2,
\end{aligned} \tag{3.34}
$$

and

$$
\begin{aligned}
|e_y^{N-2}| &\leq \mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_y^{N-1}|] + \frac{1}{2}\Delta t\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_f^{N-1}|] + \frac{1}{2}\Delta t|e_f^{N-2}| + |R_y^{N-2}| \\
&\leq C(\Delta t)^3 + \frac{L}{2}\Delta t\mathbb{E}_{t_{N-2}}^{x^{N-2}}[|e_y^{N-1}| + |e_z^{N-1}|] \\
&\qquad + \frac{L}{2}\Delta t(|e_y^{N-2}| + |e_z^{N-2}|) + C(\Delta t)^3 \\
&\leq LC(\Delta t)^3 + \frac{L}{2}\Delta t|e_y^{N-2}| + C(\Delta t)^3 \\
&\leq \frac{L}{2}\Delta t|e_y^{N-2}| + C(\Delta t)^3,
\end{aligned}
$$

which implies

$$
|e_y^{N-2}| \leq \frac{C(\Delta t)^3}{1 - L\Delta t/2} \leq C(\Delta t)^3. \tag{3.35}
$$

Similarly we can obtain the estimates

$$
|e_z^{N-3}| \leq C(\Delta t)^2, \qquad |e_y^{N-3}| \leq C(\Delta t)^3. \tag{3.36}
$$

Thus the proof is completed. □

LEMMA 3.6. *Suppose* $f \in C_b^{0,2,2}$*. Then when* Δt *is sufficiently small, it holds that for* $0 \le n \le N-3$*, if* $|e_z^{n+1}|^2 \le \Delta t$ *and* $|e_y^{n+1}|^2 \le \Delta t$*, then*

$$\begin{aligned}
&|e_y^n|^2 + |e_z^n|^2 \\
&\quad \le (1 + C\Delta t)\Big(\frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] + |\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]\Big) \\
&\qquad + C\Delta t(|e_y^n|^2 + |e_z^n|^2) \\
&\qquad + C\frac{|R_y^n|^2}{\Delta t} + C\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]|^2 + |R_z^n - \mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|^2}{(\Delta t)^3},
\end{aligned} \tag{3.37}$$

where $C > 0$ *is a generic constant depending only on* T *and the bounds of derivatives of* φ *and* f*.*

Proof. For stochastic processes X and Y, let us define

$$Var^n(X) = \mathbb{E}_{t_n}^{x^n}[|X|^2] - |\mathbb{E}_{t_n}^{x^n}[X]|^2, \quad Cov^n(X,Y) = \mathbb{E}_{t_n}^{x^n}[XY] - \mathbb{E}_{t_n}^{x^n}[X]\mathbb{E}_{t_n}^{x^n}[Y],$$

for $0 \le n \le N-1$.

By the equations (3.26) and (3.27), for $0 \le n \le N-3$, it holds that

$$\begin{aligned}
e_z^n &= \mathbb{E}_{t_n}^{x^n}[e_z^{n+2}] - \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+2}] \\
&\quad - \mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+2}] + \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+1}] \\
&\quad + \mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+1}] + \mathbb{E}_{t_n}^{x^n}[e_f^{n+1}\Delta W_{n+1}] + \mathbb{E}_{t_n}^{x^n}[e_f^{n+1}\Delta W_{n+1}] \\
&\quad + \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}] + 2\frac{R_z^n - \mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]}{\Delta t} \\
&= \mathbb{E}_{t_n}^{x^n}[e_z^{n+2}] - \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+2}] + \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+1}] \\
&\quad + \mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+1}] - \mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+2}] \\
&\quad + 2\mathbb{E}_{t_n}^{x^n}[e_f^{n+1}\Delta W_{n+1}] + \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}] + 2\frac{R_z^n - \mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]}{\Delta t}.
\end{aligned} \tag{3.38}$$

From the definition of the conditional mathematical expectation $\mathbb{E}_{t_n}^{x^n}[\cdot]$, we have

$$\begin{aligned}
&\mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}\Delta W_{n+1}] \\
&\quad = \mathbb{E}_{t_n}^{x^n}[u(t_{n+2}, x^n + W_{t_{n+2}} - W_{t_n})\Delta W_{n+1}] \\
&\quad = \mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+1}}^{x^n + W_{t_{n+1}} - W_{t_n}}[u(t_{n+2}, x^n + W_{t_{n+2}} - W_{t_{n+1}} + W_{t_{n+1}} - W_{t_n})]\Delta W_{n+1}] \\
&\quad = \Delta t\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+1}}^{x^n + W_{t_{n+1}} - W_{t_n}}[\nabla_x u(t_{n+2}, x^n + W_{t_{n+2}} - W_{t_{n+1}} + W_{t_{n+1}} - W_{t_n})]] \\
&\quad = \Delta t\mathbb{E}_{t_n}^{x^n}[\nabla y_{t_{n+2}}],
\end{aligned} \tag{3.39}$$

and

$$\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}(W_{t_{n+2}} - W_{t_n})] &= \frac{1}{\sqrt{4\pi\Delta t}}\int_{-\infty}^{\infty} u(t_{n+2}, x^n + v)v\exp(-\frac{v^2}{4\Delta t})\,dv \\
&= \frac{2\Delta t}{\sqrt{4\pi\Delta t}}\int_{-\infty}^{\infty} \nabla_x u(t_{n+2}, x^n + v)\exp(-\frac{v^2}{4\Delta t})\,dv \\
&= 2\Delta t\mathbb{E}_{t_n}^{x^n}[\nabla y_{t_{n+2}}].
\end{aligned} \tag{3.40}$$

Thus we get

$$\begin{aligned}\mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}\Delta W_{n+2}] &= \mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}(W_{t_{n+2}} - W_{t_n} + W_{t_n} - W_{t_{n+1}})] \\ &= \mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}(W_{t_{n+2}} - W_{t_n})] - \mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}(W_{t_{n+1}} - W_{t_n})] \\ &= 2\Delta t\mathbb{E}_{t_n}^{x^n}[\nabla y_{t_{n+2}}] - \Delta t\mathbb{E}_{t_n}^{x^n}[\nabla y_{t_{n+2}}] \\ &= \Delta t\mathbb{E}_{t_n}^{x^n}[\nabla y_{t_{n+2}}].\end{aligned} \tag{3.41}$$

Using (3.39) and (3.41), we obtain the identity

$$\mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}\Delta W_{n+1}] = \mathbb{E}_{t_n}^{x^n}[y_{t_{n+2}}\Delta W_{n+2}]. \tag{3.42}$$

If the Brownian motion starts from the time-space point (t_n, x^n), then y^{n+k} is a function of $x^n + W_{t_{n+k}} - W_{t_n}$, that is, y^{n+k} is a function of $x^n + W_{t_{n+k}} - W_{t_n}$. Then similar to the way to obtain (3.42), it holds

$$\mathbb{E}_{t_n}^{x^n}[y^{n+2}\Delta W_{n+1}] = \mathbb{E}_{t_n}^{x^n}[y^{n+2}\Delta W_{n+2}]. \tag{3.43}$$

Thus (3.42) and (3.43) give us the following identity

$$\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+1}] = \mathbb{E}_{t_n}^{x^n}[e_y^{n+2}\Delta W_{n+2}]. \tag{3.44}$$

Similarly we can get

$$\mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+1}] = \mathbb{E}_{t_n}^{x^n}[e_f^{n+2}\Delta W_{n+2}]. \tag{3.45}$$

Combining the equations (3.38), (3.44) and (3.45) we obtain

$$\begin{aligned}e_z^n =&\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}] + 2\mathbb{E}_{t_n}^{x^n}[e_f^{n+1}\Delta W_{n+1}] \\ &+ \frac{2}{\Delta t}\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}] + 2\frac{R_z^n - \mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]}{\Delta t}.\end{aligned} \tag{3.46}$$

Using the Taylor expansion it is easy to obtain that for $i = N-1, N-2, \ldots, 0$,

$$e_f^i = f(t_i, y_{t_i}, z_{t_i}) - f(t_i, y^i, z^i) = \xi^i e_y^i + \gamma^i e_z^i \tag{3.47}$$

where

$$\xi^i = f_y(t_i, y_{t_i} - \beta^i e_y^i, z_{t_i} - \beta^i e_z^i), \tag{3.48}$$

$$\gamma^i = f_z(t_i, y_{t_i} - \beta^i e_y^i, z_{t_i} - \beta^i e_z^i) \tag{3.49}$$

with some $\beta^i \in [0,1]$. Then by (3.46) and (3.47) we deduce

$$\begin{aligned}e_z^n = &\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}] + 2\mathbb{E}_{t_n}^{x^n}[(\xi^{n+1}e_y^{n+1} + \gamma^{n+1}e_z^{n+1})\Delta W_{n+1}] \\ &+ \frac{2\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]}{\Delta t} + 2\frac{R_z^n - \mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]}{\Delta t}.\end{aligned} \tag{3.50}$$

From the facts that

$$\begin{aligned}\mathbb{E}_{t_n}^{x^n}[f_y(t_n, y_{t_n}, z_{t_n})\Delta W_{n+1}] = f_y(t_n, y_{t_n}, z_{t_n})\mathbb{E}_{t_n}^{x^n}[\Delta W_{n+1}] = 0, \\ \mathbb{E}_{t_n}^{x^n}[f_z(t_n, y_{t_n}, z_{t_n})\Delta W_{n+1}] = f_z(t_n, y_{t_n}, z_{t_n})\mathbb{E}_{t_n}^{x^n}[\Delta W_{n+1}] = 0,\end{aligned}$$

and the definition of $Cov^n(\cdot)$, we have

$$\begin{aligned}
&\mathbb{E}_{t_n}^{x^n}[\xi^{n+1}e_y^{n+1}\Delta W_{n+1}]\\
&\quad= \mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]\mathbb{E}_{t_n}^{x^n}[\xi^{n+1}\Delta W_{n+1}] + Cov^n(e_y^{n+1}, \xi^{n+1}\Delta W_{n+1})\\
&\quad= \mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]\mathbb{E}_{t_n}^{x^n}[(\xi^{n+1} - f_y(t_n, y_{t_n}, z_{t_n}))\Delta W_{n+1}]\\
&\qquad+ Cov^n(e_y^{n+1}, \xi^{n+1}\Delta W_{n+1}),
\end{aligned} \tag{3.51}$$

and

$$\begin{aligned}
&\mathbb{E}_{t_n}^{x^n}[\gamma^{n+1}e_z^{n+1}\Delta W_{n+1}]\\
&\quad= \mathbb{E}_{t_n}^{x^n}[e_z^{n+1}]\mathbb{E}_{t_n}^{x^n}[(\gamma^{n+1} - f_z(t_n, y_{t_n}, z_{t_n}))\Delta W_{n+1}]\\
&\qquad+ Cov^n(e_z^{n+1}, \gamma^{n+1}\Delta W_{n+1}).
\end{aligned} \tag{3.52}$$

By (3.48), (3.49) and using the Taylor expansion again, we obtain

$$\begin{aligned}
&\xi^{n+1} - f_y(t_n, y_{t_n}, z_{t_n})\\
&\quad= f_y(t_{n+1}, y_{t_{n+1}}, z_{t_{n+1}}) - f_y(t_n, y_{t_n}, z_{t_n})\\
&\qquad- f_{yy}(t_{n+1}, y_{t_{n+1}} - \alpha_1^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_1^{n+1}e_z^{n+1})\beta_1^{n+1}e_y^{n+1}\\
&\qquad- f_{yz}(t_{n+1}, y_{t_{n+1}} - \alpha_2^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_2^{n+1}e_z^{n+1})\beta_1^{n+1}e_z^{n+1},
\end{aligned} \tag{3.53}$$

and

$$\begin{aligned}
&\gamma^{n+1} - f_z(t_n, y_{t_n}, z_{t_n})\\
&\quad= f_z(t_{n+1}, y_{t_{n+1}}, z_{t_{n+1}}) - f_z(t_n, y_{t_n}, z_{t_n})\\
&\qquad- f_{zy}(t_{n+1}, y_{t_{n+1}} - \alpha_3^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_3^{n+1}e_z^{n+1})\beta_2^{n+1}e_y^{n+1}\\
&\qquad- f_{zz}(t_{n+1}, y_{t_{n+1}} - \alpha_4^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_4^{n+1}e_z^{n+1})\beta_2^{n+1}e_z^{n+1},
\end{aligned} \tag{3.54}$$

where $\alpha_i^{n+1} \in [0,1]$ $(i = 1,2,3,4)$ and $\beta_i^{n+1} \in [0,1]$ $(i = 1,2)$. Using the equations (3.53) and (3.54), the Hölder inequality, and the assumption that $|e_z^{n+1}|^2 \le \Delta t$ and $|e_y^{n+1}|^2 \le \Delta t$, we obtain

$$\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[(\xi^{n+1} - f_y(t_n, y_{t_n}, z_{t_n}))\Delta W_{n+1}]|\\
&\quad= |\mathbb{E}_{t_n}^{x^n}[(f_y(t_{n+1}, y_{n+1}, z_{t_{n+1}}) - f_y(t_n, y_{t_n}, z_{t_n}))\Delta W_{n+1}]\\
&\qquad- \mathbb{E}_{t_n}^{x^n}[f_{yy}(t_{n+1}, y_{t_{n+1}} - \alpha_1^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_1^{n+1}e_z^{n+1})\beta_1^{n+1}e_y^{n+1}\Delta W_{n+1}]\\
&\qquad- \mathbb{E}_{t_n}^{x^n}[f_{yz}(t_{n+1}, y_{t_{n+1}} - \alpha_2^{n+1}e_y^{n+1}, z_{t_{n+1}} - \alpha_2^{n+1}e_z^{n+1})\beta_1^{n+1}e_z^{n+1}\Delta W_{n+1}]|\\
&\quad\le \mathbb{E}_{t_n}^{x^n}[(f_y(t_{n+1}, y_{t_{n+1}}, z_{t_{n+1}}) - f_y(t_n, y_{t_n}, z_{t_n}))^2]^{\frac{1}{2}}\mathbb{E}_{t_n}^{x^n}[(\Delta W_{n+1})^2]^{\frac{1}{2}}\\
&\qquad+ C\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2]^{\frac{1}{2}}\mathbb{E}_{t_n}^{x^n}[|\Delta W_{n+1}|^2]^{\frac{1}{2}} + C\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]^{\frac{1}{2}}\mathbb{E}_{t_n}^{x^n}[|\Delta W_{n+1}|^2]^{\frac{1}{2}}\\
&\quad\le C\Delta t + C\sqrt{\Delta t}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2]^{\frac{1}{2}} + C\sqrt{\Delta t}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]^{\frac{1}{2}}\\
&\quad\le C\Delta t.
\end{aligned} \tag{3.55}$$

Similarly we can get

$$|\mathbb{E}_{t_n}^{x^n}[(\gamma^{n+1} - f_z(t_n, y_{t_n}, z_{t_n}))\Delta W_{t_{n+1}}]| \le C\Delta t. \tag{3.56}$$

Using the Cauchy-Schwarz inequality it gives us that

$$\begin{aligned}&|Cov^n(e_y^{n+1},\xi^{n+1}\Delta W_{n+1})|\\&\quad\leq\{Var^n(e_y^{n+1})Var^n(\xi^{n+1}\Delta W_{n+1})\}^{\frac{1}{2}}\\&\quad\leq\mathbb{E}_{t_n}^{x^n}[(\xi^{n+1}\Delta W_{n+1})^2]^{\frac{1}{2}}\{Var^n(e_y^{n+1})\}^{\frac{1}{2}}\leq C\sqrt{\Delta t}\{Var^n(e_y^{n+1})\}^{\frac{1}{2}},\end{aligned}\tag{3.57}$$

and

$$|Cov^n(e_z^{n+1},\gamma^{n+1}\Delta W_{n+1})|\leq C\sqrt{\Delta t}\{Var^n(e_z^{n+1})\}^{\frac{1}{2}}.\tag{3.58}$$

By the inequalities (3.51), (3.52), (3.55), (3.56), (3.57) and (3.58) we easily obtain

$$\begin{aligned}&|\mathbb{E}_{t_n}^{x^n}[\xi^{n+1}e_y^{n+1}\Delta W_{t_{n+1}}]|\leq C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|]+C\sqrt{\Delta t}\{Var^n(e_y^{n+1})\}^{\frac{1}{2}},\\&|\mathbb{E}_{t_n}^{x^n}[\gamma^{n+1}e_z^{n+1}\Delta W_{t_{n+1}}]|\leq C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|]+C\sqrt{\Delta t}\{Var^n(e_z^{n+1})\}^{\frac{1}{2}}.\end{aligned}\tag{3.59}$$

It follows from (3.50) and (3.59) that

$$\begin{aligned}|e_z^n|\leq&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|+C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|+|e_z^{n+1}|]\\&+C\sqrt{\Delta t}\{(Var^n(e_y^{n+1}))^{\frac{1}{2}}+(Var^n(e_z^{n+1}))^{\frac{1}{2}}\}\\&+\frac{2|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]|}{\Delta t}+\frac{2|R_z^n-\mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|}{\Delta t}.\end{aligned}\tag{3.60}$$

Then by the inequality $2ab\leq\eta\Delta ta^2+\dfrac{1}{\eta\Delta t}b^2$ (for any $\eta>0$) and taking square on both sides of the inequality (3.60), we obtain

$$\begin{aligned}|e_z^n|^2\leq&(1+\eta\Delta t)|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2+\tilde{C}(1+\frac{1}{\eta\Delta t})\Big((\Delta t)^2\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2+|e_z^{n+1}|^2]\\&+\Delta tVar^n(e_y^{n+1})+\Delta tVar^n(e_z^{n+1})\\&+\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]|^2+|R_z^n-\mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|^2}{(\Delta t)^2}\Big),\end{aligned}\tag{3.61}$$

for some generic constant $\tilde{C}>0$. By choosing $\eta=2\tilde{C}$ in the inequality (3.61) we get

$$\begin{aligned}|e_z^n|^2\leq&(1+C\Delta t)|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2\\&+\frac{1}{2}(1+C\Delta t)\left(\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2+|e_z^{n+1}|^2]+Var^n(e_y^{n+1})+Var^n(e_z^{n+1})\right)\\&+\frac{1}{2}(1+C\Delta t)\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]|^2+|R_z^n-\mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|^2}{(\Delta t)^3}.\end{aligned}\tag{3.62}$$

By (3.26) we know

$$|e_y^n|\leq|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|+\frac{L}{2}\Delta t\left(|e_y^n|+|e_z^n|+\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|+|e_z^{n+1}|]\right)+|R_y^n|.\tag{3.63}$$

Taking square of the inequality (3.63) yields

$$\begin{aligned}|e_y^n|^2\leq&(1+C\Delta t)|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2+|e_z^{n+1}|^2]\\&+C\Delta t(|e_y^n|^2+|e_z^n|^2)+\frac{C|R_y^n|^2}{\Delta t}.\end{aligned}\tag{3.64}$$

From (3.62) and (3.64), we easily deduce

$$
\begin{aligned}
&|e_y^n|^2+|e_z^n|^2\\
&\quad\le(1+C\Delta t)\Big(|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2+\frac{1}{2}Var^n(e_y^{n+1})+\frac{1}{2}Var^n(e_z^{n+1})\Big)\\
&\qquad+C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2+|e_z^{n+1}|^2]+C\Delta t(|e_y^n|^2+|e_z^n|^2)\\
&\qquad+C\frac{|R_y^n|^2}{\Delta t}+C\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{n+1}]|^2+|R_z^n-\mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|^2}{(\Delta t)^3}.
\end{aligned}
\tag{3.65}
$$

Notice that

$$
|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+\frac{1}{2}Var^n(e_y^{n+1})=\frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2],
\tag{3.66}
$$

$$
Var^n(e_z^{n+1})\le\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2],
\tag{3.67}
$$

then we immediate obtain the result (3.37) by putting (3.66) and (3.67) into (3.65). The proof is completed. □

The following lemma gives us a very useful recursive relation on grow of the errors.

LEMMA 3.7. *Suppose that $f\in C_b^{3,6,6}$, and $\varphi\in C_b^{7+\alpha}$ for some $\alpha\in(0,1)$. Then when Δt is sufficiently small, it holds that for $0\le n\le N-5$, if $|e_z^k|^2\le\Delta t$ and $|e_y^k|^2\le\Delta t$ for $k=n+2,n+3$, then*

$$
\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]+\frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2]\\
&\quad\le(1+C\Delta t)\Big(\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_z^{n+4}]|^2]+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+3}|^2]\\
&\qquad+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2]+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|e_y^{n+3}|^2]]\Big)+C(\Delta t)^5,
\end{aligned}
\tag{3.68}
$$

where $C>0$ is a generic constant depending only on T and the bounds of derivatives of φ and f.

Proof. Replacing n by $n+1$ in the inequality (3.62) gives us

$$
\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]&\le(1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+3}]|^2]\\
&\quad+\frac{1+C\Delta t}{2}\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2+|e_z^{n+2}|^2]\\
&\quad+\frac{1+C\Delta t}{2}\mathbb{E}_{t_n}^{x^n}[Var^{n+1}(e_y^{n+2})+Var^{n+1}(e_z^{n+2})]\\
&\quad+C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_y^{n+2}\Delta W_{n+2}]|^2+|R_z^{n+1}-\mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_z^{n+2}]|^2]}{(\Delta t)^3},
\end{aligned}
\tag{3.69}
$$

then we have

$$
\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad \le (1+C\Delta t)\Bigg(|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+3}]|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] + \frac{1}{4}\mathbb{E}_{t_n}^{x^n}[Var^{n+1}(e_y^{n+2}) + Var^{n+1}(e_z^{n+2})]\Bigg) \\
&\qquad + \frac{1+C\Delta t}{4}\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2 + |e_z^{n+2}|^2] \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_y^{n+2}\Delta W_{n+2}]|^2 + |R_z^{n+1} - \mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_z^{n+2}]|^2]}{(\Delta t)^3}.
\end{aligned}
\tag{3.70}
$$

By (3.26) and Jensen's inequality, we deduce

$$
\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \le &(1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2] + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2 + |e_z^{n+2}|^2] \\
&+ C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2 + |e_z^{n+1}|^2] + C\frac{\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2]}{\Delta t},
\end{aligned}
\tag{3.71}
$$

which implies

$$
\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \le &(1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2] \\
&+ C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2 + |e_z^{n+2}|^2] + \frac{C\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2]}{\Delta t}.
\end{aligned}
\tag{3.72}
$$

By (3.26) we easily get

$$
\begin{aligned}
|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 \le &(1+C\Delta t)|\mathbb{E}_{t_n}^{x^n}[e_y^{n+2}]|^2 + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2 + |e_z^{n+2}|^2] \\
&+ C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2 + |e_z^{n+1}|^2] + C\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}]|^2}{\Delta t}.
\end{aligned}
\tag{3.73}
$$

Combining the inequalities (3.72) and (3.73), we then obtain

$$
\begin{aligned}
&\tfrac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \tfrac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] + \tfrac{1}{4}\mathbb{E}_{t_n}^{x^n}[Var^{n+1}(e_y^{n+2})] \\
&\quad \le (1+C\Delta t)\left(\tfrac{1}{2}|\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_y^{n+2}]|^2] + \tfrac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2]\right) \\
&\qquad + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2 + |e_z^{n+2}|^2] + C\tfrac{\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2]}{\Delta t} \\
&\qquad + \tfrac{1}{4}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+1}}^{x^{n+1}}[|e_y^{n+2}|^2] - |\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_y^{n+2}]|^2] \\
&\quad \le (1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2] + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2 + |e_z^{n+2}|^2] + C\tfrac{\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2]}{\Delta t}.
\end{aligned}
\tag{3.74}
$$

Notice that

$$
\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{4}\mathbb{E}_{t_n}^{x^n}[Var^{n+1}(e_z^{n+2})] \\
&\quad = |\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{4}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+1}}^{x^{n+1}}[|e_z^{n+2}|^2] - |\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+2}]|^2] \\
&\quad \le \mathbb{E}_{t_n}^{x^n}[|e_z^{n+2}|^2].
\end{aligned}
\tag{3.75}
$$

Using (3.70), (3.74) and (3.75) we then deduce

$$
\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad \le (1+C\Delta t)\left(\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2 + |e_z^{n+2}|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+3}]|^2]\right) \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_y^{n+2}\Delta W_{n+2}]|^2 + |R_z^{n+1} - \mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_z^{n+2}]|^2]}{(\Delta t)^3} \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2]}{\Delta t}.
\end{aligned} \tag{3.76}
$$

Replacing n by $n+2$ in (3.61) and (3.64) gives us

$$
\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[|e_y^{n+2}|^2] \le{}& (1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+3}|^2 + |e_z^{n+3}|^2] \\
&+ C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_z^{n+2}|^2] + \frac{C\mathbb{E}_{t_n}^{x^n}[|R_y^{n+2}|^2]}{\Delta t},
\end{aligned} \tag{3.77}
$$

and

$$
\begin{aligned}
\mathbb{E}_{t_n}^{x^n}[|e_z^{n+2}|^2] \le{}& (1+C\Delta t)\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_z^{n+4}]|^2] \\
&+ \frac{1+C\Delta t}{2}\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+3}|^2 + |e_z^{n+3}|^2] \\
&+ \frac{1+C\Delta t}{2}\mathbb{E}_{t_n}^{x^n}[Var^{n+2}(e_y^{n+3}) + Var^{n+2}(e_z^{n+3})] \\
&+ C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[R_y^{n+3}\Delta W_{n+3}]|^2 + |R_z^{n+2} - \mathbb{E}_{t_{n+2}}^{x^{n+2}}[R_z^{n+3}]|^2]}{(\Delta t)^3}.
\end{aligned} \tag{3.78}
$$

Notice that

$$
\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+3}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[Var^{n+2}(e_z^{n+3})] \le \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+3}|^2]
$$

and

$$
\begin{aligned}
&\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[Var^{n+2}(e_y^{n+3})] \\
&\qquad = \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|e_y^{n+3}|^2]].
\end{aligned}
$$

Then by using (3.76), (3.77), (3.78), Lemma 3.3, 3.4 and above identities, we have

$$
\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad\le (1+C\Delta t)\Big(\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + \mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_z^{n+4}]|^2] \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[Var^{n+2}(e_y^{n+3}) + Var^{n+2}(e_z^{n+3})] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[e_z^{n+3}]|^2]\Big) \\
&\qquad + C\Delta t\mathbb{E}_{t_n}^{x^n}[|e_y^{n+3}|^2 + |e_z^{n+3}|^2] \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_y^{n+2}\Delta W_{n+2}]|^2 + |R_z^{n+1} - \mathbb{E}_{t_{n+1}}^{x^{n+1}}[R_z^{n+2}]|^2]}{(\Delta t)^3} \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|R_y^{n+1}|^2 + |R_y^{n+2}|^2]}{\Delta t} \\
&\qquad + C\frac{\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[R_y^{n+3}\Delta W_{n+3}]|^2 + |R_z^{n+2} - \mathbb{E}_{t_{n+2}}^{x^{n+2}}[R_z^{n+3}]|^2]}{(\Delta t)^3} \\
&\quad\le (1+C\Delta t)\Big(\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_z^{n+4}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+3}|^2] \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|e_y^{n+3}|^2]]\Big) + C(\Delta t)^5.
\end{aligned}
\tag{3.79}
$$

The proof is then completed. □

Now let us present the main result on convergence of the Crank-Nicolson scheme for solving BSDEs.

THEOREM 3.8. *Let (y_t, z_t) be the solution of the BSDE (1.2) with the terminal condition $y_T = \varphi(W_T)$, and (y^n, z^n) be its approximate solution produced by using the Crank-Nicolson scheme. Suppose that $f \in C_b^{3,6,6}$, and $\varphi \in C_b^{7+\alpha}$ for some $\alpha \in (0,1)$, and the initial error satisfies*

$$
\mathbb{E}_{t_{N-1}}^{x^{N-1}}[|y_{t_N} - y^N|^2] \le C(\Delta t_{N-1})^3
$$

for some constant $C > 0$. Then we have the following L^2 error estimate: when Δt is sufficiently small, it holds that for $0 \le n \le N-1$,

$$
\max_{0\le l\le n\le N-1} \mathbb{E}_{t_l}^{x_l}[|y_{t_n} - y^n|^2 + |z_{t_n} - z^n|^2] \le C_{T,\varphi,f}(\Delta t)^4, \tag{3.80}
$$

where $C_{T,\varphi,f} > 0$ is a generic constant depending only on T and the bounds of derivatives of φ and f.

Proof. Let us first choose constants $C_* > 0$ and $1 > \delta_* > 0$ such that Lemmas 3.3-3.7 hold for this C_* when the time step size $\Delta t \le \delta_*$. Set $C_s = C_* e^{C_* T}(3+T)$. Then we define some constants in the below:

$$
C_{T,\varphi,f} = 3C_s + 2C_*^2 + 4C_*^3,
$$

$$
\delta = min\{\delta_*, \frac{1}{2C_*}, \frac{1}{C_{T,\varphi,f}}\}.
$$

In the following, we will show by *induction* that that for any k with $0 \le k \le N-1$, it holds that

$$
|e_y^k|^2 + |e_z^k|^2 \le C_{T,\varphi,f}(\Delta t)^4, \tag{3.81}
$$

when the time step size $\Delta t < \delta$. We would like to point out that (3.81) also implies

$$|e_y^k|^2 \leq \Delta t, \qquad |e_z^k|^2 \leq \Delta t, \tag{3.82}$$

since $C_{T,\varphi,f}(\Delta t)^4 \leq C_{T,\varphi,f}(1/C_{T,\varphi,f})(\Delta t)^3 \leq (\Delta t)^3 \leq \Delta t$.

First by Lemma 3.5, we know (3.81) clearly holds for $k = N-1, N-2, N-3$ since $C_{T,\varphi,f} \geq C_*$ and $\delta \leq \delta_*$.

Now we assume that (3.81) holds for all $n+1 \leq k \leq N-1$. Note that it also means (3.82) holds for all $n+1 \leq k \leq N-1$. We will show that (3.81) also hold for $k = n$ under such assumption.

Using Lemma 3.7 we get

$$\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad \leq (1 + C_*\Delta t)\big(\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_z^{n+4}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+3}|^2] \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{n+2}}^{x^{n+2}}[e_y^{n+3}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|e_y^{n+3}|^2]]\Big) + C_*(\Delta t)^5 \\
&\quad \leq (1 + C_*\Delta t)^2\big(\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|\mathbb{E}_{t_{n+4}}^{x^{n+4}}[e_z^{n+6}]|^2] + \frac{1}{2}\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|e_z^{n+5}|^2] \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_{n+2}}^{x^{n+2}}[|\mathbb{E}_{t_{n+4}}^{x^{n+4}}[e_y^{n+5}]|^2] + \frac{1}{2}\mathbb{E}_{t_{n+2}}^{x^{n+2}}[\mathbb{E}_{t_{n+4}}^{x^{n+4}}[|e_y^{n+5}|^2]]\Big) \\
&\qquad + [C_* + (1 + C_*\Delta t)C_*](\Delta t)^5.
\end{aligned} \tag{3.83}$$

If $N-n-3$ is even, by repeating the above process we get

$$\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad \leq (1 + C_*\Delta t)^{\frac{N-n-3}{2}}\Big(\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{N-3}}^{x^{N-3}}[e_z^{N-1}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{N-2}|^2] \\
&\qquad + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|\mathbb{E}_{t_{N-3}}^{x^{N-3}}[e_y^{N-2}]|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[\mathbb{E}_{t_{N-3}}^{x^{N-3}}[|e_y^{N-2}|^2]]\Big) \\
&\qquad + C_*(\Delta t)^5 \sum_{i=1}^{\frac{N-n-3}{2}} (1 + C_*\Delta t)^{i-1} \\
&\quad \leq e^{C_*(N-3)\Delta t/2}\Big(\mathbb{E}_{t_n}^{x^n}[|e_z^{N-1}|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{N-2}|^2] + \mathbb{E}_{t_n}^{x^n}[|e_y^{N-2}|^2]\Big) \\
&\qquad + C_*(\Delta t)^4((N-3)\Delta t)e^{C_*(N-3)\Delta t/2} \\
&\quad \leq 3C_*e^{C_*T}(\Delta t)^4 + C_*Te^{C_*T}(\Delta t)^4 \\
&\quad = C_s(\Delta t)^4.
\end{aligned} \tag{3.84}$$

Similarly, if $N-n-3$ is odd, then we obtain

$$\begin{aligned}
&|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2] + \frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2 + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2] \\
&\quad \leq e^{C_*(N-4)\Delta t/2}\Big(\mathbb{E}_{t_n}^{x^n}[|e_z^{N-2}|^2] + \frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{N-3}|^2] + \mathbb{E}_{t_n}^{x^n}[|e_y^{N-3}|^2]\Big) \\
&\qquad + C_*(\Delta t)^4((N-4)\Delta t/2)e^{C_*(N-3)\Delta t/2} \\
&\quad \leq 3C_*e^{C_*T}(\Delta t)^4 + C_*Te^{C_*T}(\Delta t)^4 \\
&\quad = C_s(\Delta t)^4.
\end{aligned} \tag{3.85}$$

Now by the above results and Lemmas 3.3, 3.4 and 3.6, we have

$$\begin{aligned}|e_y^n|^2+|e_z^n|^2 \le& \frac{1+C_*\Delta t}{1-C_*\Delta t}\Big(\frac{1}{2}|\mathbb{E}_{t_n}^{x^n}[e_y^{n+1}]|^2+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_y^{n+1}|^2]+|\mathbb{E}_{t_n}^{x^n}[e_z^{n+2}]|^2\\ &+\frac{1}{2}\mathbb{E}_{t_n}^{x^n}[|e_z^{n+1}|^2]\Big)+\frac{C_*}{1-C_*\Delta t}\frac{|R_y^n|^2}{\Delta t}\\ &+\frac{C_*}{1-C_*\Delta t}\frac{|\mathbb{E}_{t_n}^{x^n}[R_y^{n+1}\Delta W_{t_{n+1}}]|^2+|R_z^n-\mathbb{E}_{t_n}^{x^n}[R_z^{n+1}]|^2}{(\Delta t)^3}\\ \le& 3C_s(\Delta t)^4+2C_*^2(\Delta t)^5+4C_*^3(\Delta t)^5\\ \le& C_{T,\varphi,f}(\Delta t)^4.\end{aligned} \tag{3.86}$$

Thus we prove (3.81) by induction.

For any integer l with $0\le l\le n$, taking the conditional mathematical expectation $\mathbb{E}_{t_l}^{x_l}[\cdot]$ for on both sides of (3.81), we immediately obtain

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^2+|e_z^n|^2]\le C_{T,\varphi,f}(\Delta t)^4, \tag{3.87}$$

for $n=N-1,\dots,0$. The proof is completed. □

Theorem 3.9. *Under the conditions of Theorem 3.8, we have the following L^p $(p\ge 1)$ error estimate: when Δt is sufficiently small, it holds that for $0\le n\le N-1$,*

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|y_{t_n}-y^n|^p+|z_{t_n}-z^n|^p]\le C_{p,T,\varphi,f}(\Delta t)^{2p}, \tag{3.88}$$

where $C_{p,T,\varphi,f}>0$ is a generic constant depending only on p, T, and the bounds of derivatives of φ and f.

Proof. There are two cases to discuss.

Case I: $p\ge 2$. For any $0\le n\le N-1$, taking the power of $\frac{p}{2}$ on both sides of the inequality (3.81) of Theorem 3.8 gives us

$$|e_y^n|^p+|e_z^n|^p\le(|e_y^n|^2+|e_z^n|^2)^{\frac{p}{2}}\le C_{p,T,\varphi,f}(\Delta t)^{2p}, \tag{3.89}$$

where $C_{p,T,\varphi,f}=(C_{T,\varphi,f})^{\frac{p}{2}}$. Thus we obtain

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^p+|e_z^n|^p]\le C_{p,T,\varphi,f}(\Delta t)^{2p}. \tag{3.90}$$

Case II: $1\le p<2$. By the Jensen inequality and the inequality (3.87) of Theorem 3.8 we have

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^p]^{\frac{2}{p}}\le \max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^{p\frac{2}{p}}]=\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^2]\le C_{T,\varphi,f}(\Delta t)^4,$$

which implies

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^p]\le(C_{T,\varphi,f})^{\frac{p}{2}}(\Delta t)^{2p}. \tag{3.91}$$

Similarly we get

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_z^n|^p]\le(C_{T,\varphi,f})^{\frac{p}{2}}(\Delta t)^{2p}. \tag{3.92}$$

Combination of the inequalities (3.91) and (3.92) leads to

$$\max_{0\le l\le n\le N-1}\mathbb{E}_{t_l}^{x_l}[|e_y^n|^p+|e_z^n|^p]\le C_{p,T,\varphi,f}(\Delta t)^{2p}, \tag{3.93}$$

where $C_{p,T,\varphi,f}=2(C_{T,\varphi,f})^{\frac{p}{2}}$. □

4. Conclusions. In this paper, we study error estimates of a special case of the θ-scheme – the Crank-Nicolson scheme for solving backward stochastic differential equations. Although the C-N scheme was numerically demonstrated to be second-order accurate [26], theoretical proof for such optimal convergence for general BSDEs remains open till now. We rigorously show that under certain reasonable regularity conditions on the generator f and the terminal condition y_T, this scheme is of second-order accuracy for solving both y_t and z_t if the errors are measured in the L^p $(p \geq 1)$ norm. A key idea in the proof is to use cancelation of local errors between successive time levels. Some of future work includes application of the techniques developed in this paper to error estimates of some other accurate schemes for solving BSDEs such as the multi-step scheme proposed in [28], and design and analysis of high-order schemes for solving FBSDEs.

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